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RESEARCH ARTICLE

Eccentric contraction increases hydrogen peroxide levels and alters gene expression through Nox2 in skeletal muscle of male mice

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Abstract

Hydrogen peroxide (H_2O_2) is one of the key signaling factors regulating skeletal muscle adaptation to muscle contractions. Eccentric (ECC) and concentric (CONC) contractions drive different muscle adaptations with ECC resulting in greater changes. The present investigation tested the hypothesis that ECC produces higher cytosolic and mitochondrial H_2O_2 concentrations [H_2O_2] and alters gene expression more than CONC. Cytosolic and mitochondrial H_2O_2-sensitive fluorescent proteins, HyPer7 and MLS-HyPer7, were expressed in the anterior tibialis muscle of C57BL6J male mice. Before and for 60 min after either CONC or ECC (100 Hz, 50 contractions), $[H_2O_2]_{cyto}$ and $[H_2O_2]_{mito}$ were measured by in vivo fluorescence microscopy. RNA sequencing was performed in control (non-contracted), CONC, and ECC muscles to identify genes impacted by the contractions. $[H_2O_2]_{cyto}$ immediately after ECC was greater than after CONC (CONC: + 6%, ECC: + 11% vs. rest, $P < 0.05$) and remained higher for at least 60 min into recovery. In contrast, the elevation of $[H_2O_2]_{mito}$ was independent of the contraction modes (time; $P < 0.0042$, contraction mode; $P = 0.4965$). The impact of ECC on $[H_2O_2]_{cyto}$ was abolished by NADPH oxidase 2 (Nox2) inhibition (GSK2795039). Differentially expressed genes were not present after CONC or ECC + GSK but were found after ECC and were enriched for vascular development and apoptosis-related genes, among others. In conclusion, in mouse anterior tibialis, ECC, but not CONC, evokes a pronounced cytosolic H_2O_2 response, caused by Nox2, that is mechanistically linked to gene expression modifications.

NEW & NOTEWORTHY This in vivo model successfully characterized the effects of eccentric (ECC) and concentric (CONC) contractions on cytosolic and mitochondrial [H_2O_2] in mouse skeletal muscle. Compared with CONC, ECC induced higher and more sustained $[H_2O_2]_{cyto}$—an effect that was abolished by Nox2 inhibition. ECC-induced $[H_2O_2]_{cyto}$ elevations were requisite for altered gene expression.

concentric contraction; cytosolic; mitochondria; reactive oxygen species (ROS); transcriptomics

INTRODUCTION

Reactive oxygen species (ROS) are important regulators of skeletal muscle plasticity and adaptability (1–4). ROS is a generic term for highly reactive molecular species derived from oxygen, including superoxide ($O_2^{\bullet-}$), hydrogen peroxide (H_2O_2), and hydroxyl radicals ($^{\bullet}OH$). H_2O_2 is produced when $O_2^{\bullet-}$ is broken down by the superoxide dismutase (SOD), which consists of three isoforms (SOD1, SOD2, and SOD3) (5). H_2O_2 produced is degraded by antioxidant enzymes such as catalase, glutathione peroxidases, and peroxiredoxins (Prxs). H_2O_2 plays an important role in redox signaling due to its high selectivity and specificity compared with $O_2^{\bullet-}$ and $^{\bullet}OH$ (6). H_2O_2 is therefore considered to be a key regulator of skeletal muscle function and adaptation. The recently proposed concept of H_2O_2-mediated signaling in skeletal muscle is a redox relay in which H_2O_2 transmits redox signals to target proteins via Prxs (7).

The major sources of H_2O_2 production in skeletal muscle have been attributed to mitochondrial metabolism and NADPH and xanthine oxidases in the cytosol and vascular endothelium (8, 9). Traditionally, increased mitochondria-derived ROS, generated by the increased oxygen consumption, have been considered the predominant source of elevated ROS production by contracting skeletal muscle (10). Recently, however, mitochondrial production of $O_2^{\bullet-}$ and H_2O_2 has been demonstrated to decrease during (11, 12), and increase gradually after, muscle contractions (13). Rodent studies have provided evidence that the major source of $O_2^{\bullet-}$ and H_2O_2 production during muscle contractions is NADPH

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oxidase 2 (Nox2) localized to the sarcolemma and T-tubules (14, 15).

In human and rodent studies, the contraction-induced increase in intramyocyte ROS acts as a signal to activate transcription factors and gene expression, regulating both acute responses and long-term training adaptations (16–19). Recently, it has become clear that compartment-specific changes in H_2O_2 are important for the regulation of contraction-induced muscle adaptations such as gene expression and metabolic regulation (8, 20). For example, cytosolic H_2O_2, which is increased via Nox2 during moderate-intensity muscle contractions, is required for muscle glucose uptake (12). Consequently, suppression of increased cytosolic Nox2-derived ROS including H_2O_2 during endurance exercise in mice suppressed redox-sensitive signaling pathways and attenuated gene expression of antioxidant enzymes (17). Nox2-derived H_2O_2 is also important for mitochondrial adaptations induced by high-intensity interval training (HIIT) (14). Interestingly, however, mitochondria-specific inhibition of ROS enhances increases in mitochondria-associated proteins induced by HIIT (21). These results highlight the importance of resolving H_2O_2 in both the cytosol and mitochondrial compartments. Unfortunately, to date, the impact of compartment-specific H_2O_2 changes on gene expression, one of the molecular mechanisms for muscle adaptations, remains unknown.

Compared with concentric contractions (CONC), eccentric contractions (ECC) can induce profound muscle damage (22, 23). Antioxidants inhibit ECC-induced muscle damage to a degree dependent upon their concentration and specific biochemistry in vivo (24–26). For example, N-acetyl cysteine reduced plasma creatine kinase concentrations following intense ECC of human vastus lateralis muscles indicative of reduced muscle damage (27), supporting that increased skeletal muscle ROS concentration contributes to ECC-induced muscle damage. In mice, oxidative stress markers in skeletal muscle are persistently elevated after ECC but not CONC (28). These findings implicate ROS, especially H_2O_2 involvement in the skeletal muscle ECC adaptive response. However, the degree to which H_2O_2 is altered in myocytes by ECC and whether such a response might be compartment-specific await resolution. To address these questions, it is necessary to determine the difference between CONC- and ECC-induced H_2O_2 responses on a compartmentalized basis and document subsequent adaptive responses, most notably as regards gene expression.

We tested the following original hypotheses in healthy mouse tibialis anterior (TA) muscle. *1*) ECC will increase $[H_2O_2]_{cyto}$ and $[H_2O_2]_{mito}$ to a greater extent than CONC. *2*) The increase in $[H_2O_2]_{cyto}$ associated with ECC contraction can be suppressed by Nox2 inhibition. *3*) Differences in $[H_2O_2]_{cyto}$ and $[H_2O_2]_{mito}$ due to contrasting contraction modes and Nox2 inhibition induce differential gene expression.

METHODS

Animals

All procedures in this study were approved by University of Electro-Communications Institutional Animal Care and Use Committee (No. A-30-2, G18001). Male C57BL6 mice (42 total, 10–16 wk of age, Japan SLC, Shizuoka, Japan) were housed at controlled temperature (25°C) and a 12:12-h light: dark cycle with food and water provided ad libitum (Rodent Lab Diet EQ; 5L37, PMI Nutrition International, St. Louis, MO). The mice were anesthetized by intraperitoneal injection of a triple combined anesthetic (0.3 mg/kg of medetomidine, 4.0 mg/kg of midazolam, and 5.0 mg/kg of butorphanol) (29), and supplemental doses of anesthesia were administered as needed. At the end of experimental protocols, animals were killed by isoflurane (Pfizer Japan, Tokyo, Japan) overdose.

Muscle Preparation for In Vivo Cytosolic and Mitochondrial H_2O_2 Imaging

About 1 wk before in vivo H_2O_2 imaging, mouse lower legs were dewormed and skin disinfected with 70% ethanol under isoflurane anesthesia (2% isoflurane-O_2 mixture). Hyaluronidase (FUJIFILM Wako Pure Chemical Corporation, Tokyo, Japan) at 40 IU/μL dissolved in sterile saline (0.36 mg/mL) was injected through the skin into the TA muscle by syringe. After 60 min, a total of 40 μL of either pCS2 + HyPer7 (No. 136466; Addgene, Watertown, MA), a plasmid for H_2O_2-sensitive fluorescent protein, or pCS2 + MLS-HyPer7 (No. 136470; Addgene, Watertown, MA), a plasmid for mitochondrial-matrix-H_2O_2-sensitive fluorescent protein, was injected into the TA muscle. Thereafter, electroporation was applied with a 5-mm S-tweezers circular platinum electrode (CUY650P5; Nepa Gene, Tokyo, Japan) connected to an electroporator (NEPA21 Type II; Nepa Gene, Tokyo, Japan). Electroporation was performed by applying three square-wave electrical pulses (30 ms) (for poring) at 50-ms intervals (80 V, 72 V, and 65 V, respectively). Subsequently, three additional pulses were applied (10 V, 6 V, and 3.6 V, respectively) at 50-ms intervals, followed by three more pulses under the same conditions except that the polarity was reversed (for transfer). In mice transfected with HyPer7 or MLS-HyPer7, TA muscle was exposed as previously reported (30). HyPer7 and MLS-HyPer7 were expressed in the muscles of different mice (i.e., both were not expressed in the same individuals).

Fluorescence image acquisition and analysis were performed using previously reported procedures (30). Specifically, 420 nm and 490 nm wavelength excitation light was delivered using a Xenon lamp equipped with appropriate bandpass filters, and pairs of fluorescence images were captured through the 516-nm emission wavelength filter for ratiometry. The 490/420-nm ratio images were used to estimate H_2O_2 concentrations. All image analyses were performed using capture software (NIS-Elements Advanced Research; Nikon, Tokyo). HyPer7 or MLS-HyPer7 expressing fibers were defined as fibers with an absolute fluorescence count of 3,000 (arbitrary units). We report that the ratio values (490 nm/420 nm) of HyPer7 expressed in the present procedure are almost constant under in vivo resting conditions for 30 min and are increased by exogenous H_2O_2 (1.0 mM concentration) treatment (30; Supplemental Figs. S1 and S2 therein). Similar to the HyPer7-expressed fibers, MLS-HyPer7-expressed fibers showed no change in ratio values after 30 min of rest (Supplemental Fig. S1; all Supplemental material is available at https://doi.org/10.6084/m9.figshare.26156413). Furthermore, an increase in ratio

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values (increased fluorescence at 490 nm and decreased fluorescence at 420 nm) was observed under exogenous H_2O_2 (1.0 mM concentration) loading (Supplemental Fig. S2). All fibers expressing HyPer7 and MLS-HyPer7 in the microscopic field of view were included in the measurements. ROIs for H_2O_2 measurements were identified along the length of the muscle fibers, and rectangles were set up on an average of nine muscle fibers per mouse. The average of the ROIs was treated as the data for that individual. For example, for an individual with five ROIs identified, the average of the five is the individual animal data for that mouse. Ratio values indicating changes in H_2O_2 levels were defined and used as follows, Rf: fluorescence ratio (=F490/F420), R_{pre}: Rf before contraction, R: relative change in Rf (=Rf/R_{pre}).

Muscle Contraction

In vivo concentric contractions group.

The TA distal tendon was connected by a fine wire to a force transducer (Model 1030, UFI, Morro Bay, CA). For the induction of TA muscle contractions, stainless steel electrodes (TF210-104C, Unique Medical, Tokyo, Japan) were attached to the peroneal nerve. After a 10- to 15-min stabilization period following the surgery, tetanic (100 Hz, 5–8 V, stimulus duration 700 ms, every 3 s for 2.5 min, i.e., 50 contractions) muscle contractions were elicited. The muscle tension development data were acquired by microcomputer using data acquisition software (Chart V. 3.6.3) at a sampling frequency of 10 kHz. The time course of $[H_2O_2]_{cyto}$ or $[H_2O_2]_{mito}$ changes was observed in response to one set of concentric muscle contractions beginning 5 min before and continuing for 60 min after contractions. One image per minute was acquired.

In vivo eccentric contractions group.

The time course of $[H_2O_2]_{cyto}$ or $[H_2O_2]_{mito}$ changes was observed in the same manner as for concentric contractions. The exercise load device (RU-72 Model: Motomura Systems, Tokyo, Japan) was linked to the force transducer by a wire attached to the TA tendon and glued to the force transducer. The TA was pulled to provide a stretch of ~19% of the resting muscle length (15-degree rotation by motor). As for CONC, after a 10- to 15-min stabilization period following the surgery, tetanic (100 Hz, 5–8 V, stimulus duration 700 ms, every 3 s for 2.5 min, i.e., 50 contractions) muscle contractions were elicited with muscle lengthening started 0.2 s after the initiation of electrical stimulation via peroneal nerve stimulation.

GSK2795039 (GSK, GlaxoSmithKline, MedChemExpress, Monmouth Junction, NJ) decreases ROS production by specifically inhibiting Nox2. Unlike other NADPH oxidase inhibitors, GSK is a small molecule ($C_{23}H_{26}N_6O_2S$, 450.56 amu) that selectively inhibits Nox2 (31). GSK was dissolved in DMSO, PEG300, Tween 80, and Saline at 5 mM and diluted to 100 μM with Krebs–Hensleit bicarbonate-buffered solution (KHB) according to the manufacturer's specifications. This concentration completely inhibits Nox2 in leukemic HL60 cells and human peripheral blood mononuclear cells (PBMCs) and reduces ROS production (31). It has also been reported that intraperitoneal administration in vivo decreases oxidative stress in myocardium (32) and partially inhibits post-ECC damage of single muscle fibers in mdx mice (33). The TA muscles were partially exposed by a surgical skin incision and incubated in GSK/KHB solution for 20 min before ECC on a 37°C hotplate. This treatment was applied only to the ECC + GSK group.

Modeling of H_2O_2 Profiles

Curve fitting to examine cytosolic H_2O_2 profiles after muscle contractions was performed using KaleidaGraph software (v. 5.0, Synergy software, Reading, PA) with a one-component model for the H_2O_2 data:

$$R(t) = R0 \text{ at } t < TD$$

$$R(t) = R0 + \Delta R[1 - e^{-(t-TD)/\tau}] \text{ at } t \geq TD$$

where R (t) is a given time point, R0 is R at time 0 (immediately after contraction), and ΔR is the amplitude of the decrease in R (i.e., 60 min after contractions). TD is the time delay and τ is the time constant. Pilot studies demonstrated that the recovery response to contractions was quasi-exponential.

RNA Sequencing

Preparation of muscle samples for RNA sequencing data.

TA muscles of 10-wk-old male mice ($n = 12$) were stimulated with the concentric/eccentric/eccentric + Nox2 inhibitor contractions (CONC/ECC/ECC + GSK) protocols shown in *Muscle Contraction* section (100 Hz, 5–9 V, stimulus duration 700 ms, every 3 s for 2.5 min i.e., 50 contractions). The control group underwent operation only. Muscles were harvested 60 min after the muscle contractions were completed and recovery measurements had been made. All muscle samples were immediately frozen in liquid nitrogen and kept at −80°C. After muscle collection, animals were euthanized by an overdose of isoflurane.

RNA extraction. TA (~50 mg) was homogenized in cold TRIzol reagent (15596018, Thermo Fisher Scientific, Waltham, MA) using a bead crusher (μT-01, Taitec, Saitama, Japan). After homogenization, total RNA was extracted using NucleoSpin RNA kit (740955, Takara, Shiga, Japan), and RNA concentration was measured using NanoDrop Lite (Thermo Fisher Scientific).

Generation and preprocessing of RNA sequencing data.

Nucleic acid quality confirmation, library preparation, and next-generation sequencing were performed by Rhelixa (Tokyo, Japan). Poly(A) selection was used to enrich mRNA from total RNA, and mRNA sample quality was verified using a 2100 Bioanalyzer (Agilent). mRNA integrity number (RIN) values ranged from 8.7 to 9.4 for the entire sample. The quality of raw paired-end sequence reads was assessed with FastQC (v. 0.11.7; https://www.bioinformatics.babraham.ac.uk/projects/fastqc/). Low-quality (<20) base and adapter sequences were trimmed by Trimmomatic software (v. 0.38) using the following parameters: ILLUMINACLIP:path/to/adapter.fa:2:30:10 LEADING:20 TRAILING:20 SLIDINGWINDOW:4:15 MINLEN:36. Trimmed reads were aligned to the reference genome using the RNA sequencing (RNA-Seq) aligner HISAT2 (v. 2.1.0). The .sam files generated by HISAT2 were converted to .bam files using Samtools (v. 1.9). The .bam files were used

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to estimate the amount of uniquely mapped reads using featureCounts (v. 1.6.3). Raw read counts were normalized by transcripts per million (TPM). Genes above TPM 1 were used for analysis.

Differential expression analysis.
Differential gene expression analysis was performed using Python. A *t* test was used to identify significantly differentially expressed genes (DEGs) between each contraction model and the control group. The Benjamini–Hochberg (BH) procedure was used to control the false discovery rate for genes with $P < 0.05$ by *t* test. In accordance with previous studies, genes with $q < 0.20$ in the BH were defined as differentially expressed (34).

Functional annotation of differentially expressed genes.
Functional features of DEGs were identified by testing the gene list for Gene Ontology terms (GO) enrichment; all three GO categories (biological process, molecular function, and cellular component) were considered during the analysis. Enriched GO terms were defined as BH correction $q < 0.05$ enriched in at least two genes (35, 36).

Statistical Analysis

Values are expressed as means (± SE). Statistical analyses were performed in Prism 9.0 (GraphPad Software, San Diego, CA). The change in exerted tension for each contraction number between conditions (CONC, ECC, and ECC + GSK; Fig. 1), time course changes in $[H_2O_2]_{cyto}$ and $[H_2O_2]_{mito}$ in each contraction mode (Fig. 2), and in the Nox2 inhibition condition (Fig. 4) was compared by two-way ANOVA, followed by Tukey's multiple comparison test. The changes in the quantitative kinetics analysis parameters of $[H_2O_2]_{cyto}$ (Fig. 3) were compared using a two-tailed unpaired Student's *t* test. The level of significance was set at $P < 0.05$.

RESULTS

Muscle Force

Figure 1*B* shows the change in tension for the contraction models used in this study. The ECC model developed higher absolute tension than the CONC model, as shown in previous studies (22, 37). The ECC + GSK condition did not demonstrate any tension difference compared with ECC.

In Vivo Cytosolic H_2O_2 Concentration and Mitochondria H_2O_2 Concentration Profiles after Concentric Contractions and Eccentric Contractions

Figure 2 shows the profiles of cytosolic H_2O_2 concentration ($[H_2O_2]_{cyto}$) and mitochondria H_2O_2 concentration ($[H_2O_2]_{mito}$). The $[H_2O_2]_{cyto}$ profiles (Fig. 2, *top*) demonstrated a significant

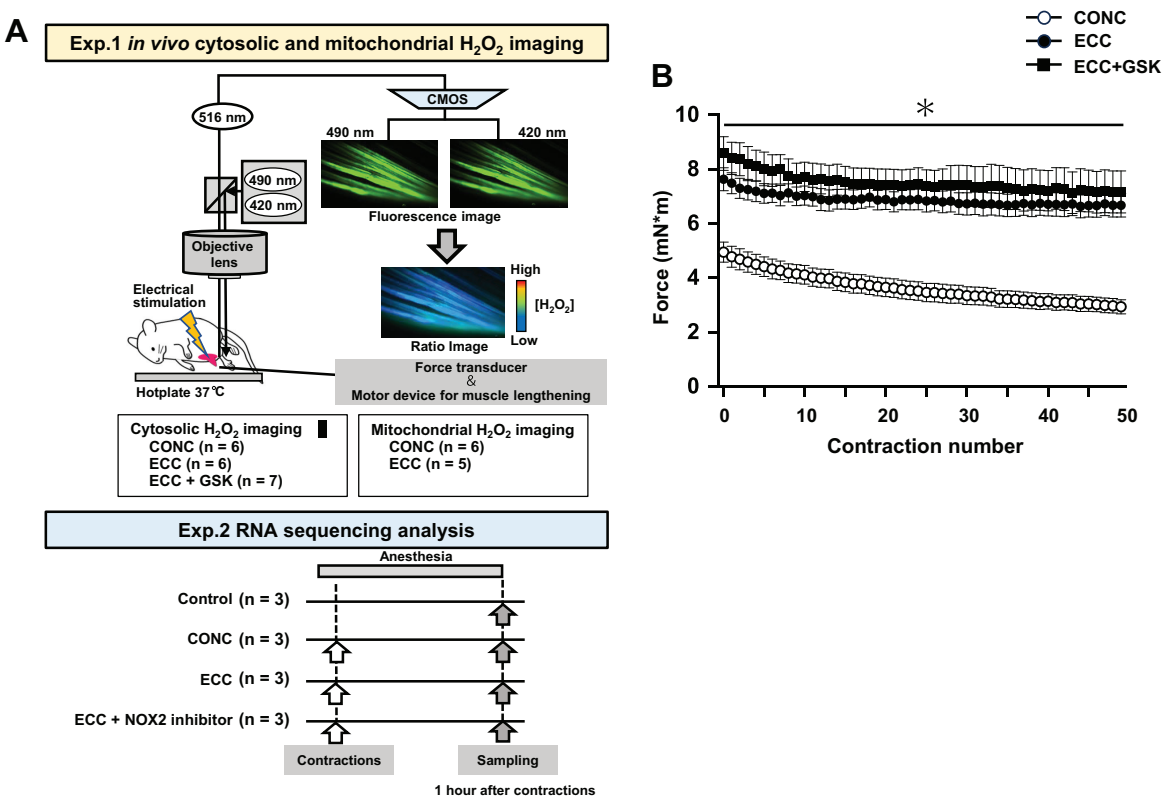


Figure 1. Schematics showing experimental procedure used to evaluate in vivo cytosolic and mitochondrial H_2O_2 dynamics (Exp.1) and RNA-sequencing analysis procedure (Exp.2). *A*: under anesthesia, the tendons of the tibialis anterior (TA) muscle of male mice were connected to a force transducer and motor device. Electrodes were mounted on the peroneal nerve. Fluorescence images were captured, and ratio (Rf: F490/F420) image was converted. *B*: muscle force profile during 50 contractions in TA muscles [concentric contractions (CONC), ○; eccentric contractions (ECC), ●; ECC + GSK (NOX2 inhibitor), ■]. Values are means ± SE. *Significant difference CONC vs. ECC and ECC + GSK for the same time points ($P < 0.05$). *n* = number of male mice. H_2O_2, hydrogen peroxide.

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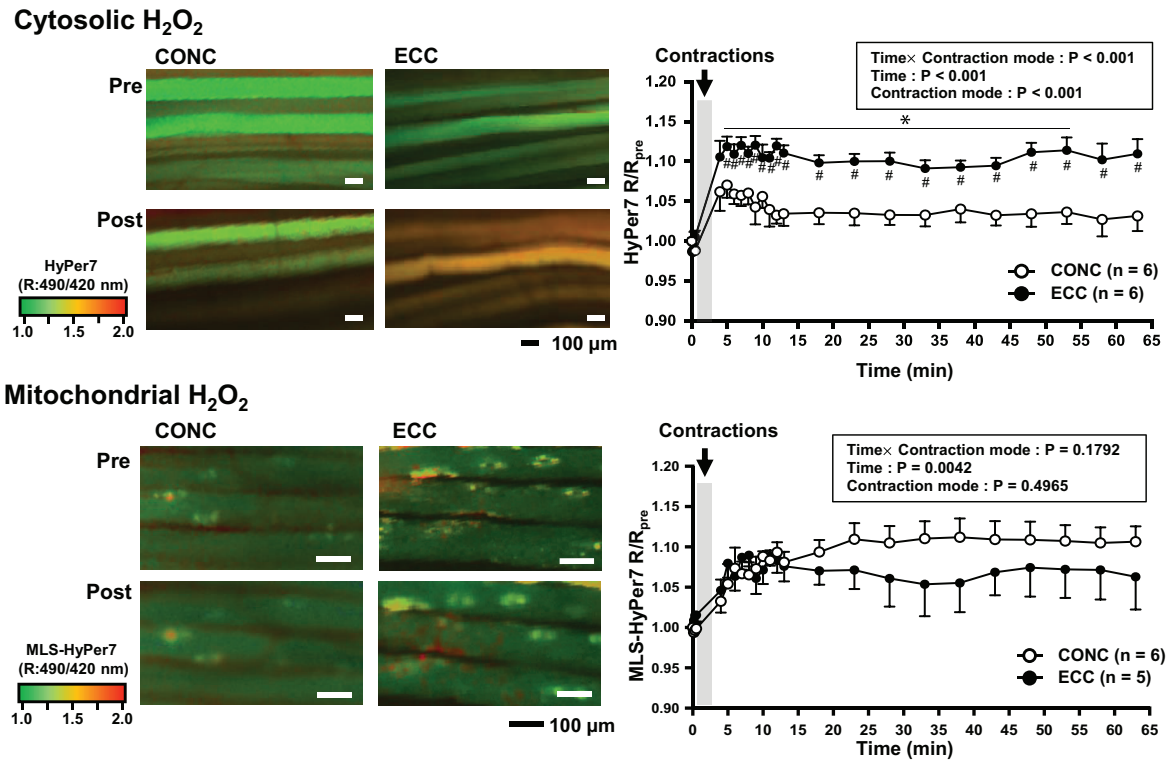


Figure 2. In vivo cytosolic hydrogen peroxide (H_2O_2) concentration ($[H_2O_2]_{cyto}$) and mitochondria H_2O_2 concentration ($[H_2O_2]_{mito}$) profiles after concentric contractions (CONC) and eccentric contractions (ECC). Typical HyPer7 (cytosolic H_2O_2) and MLS-HyPer7 (mitochondrial H_2O_2) fluorescence images were shown before and after CONC and ECC. Time-dependent change in the ratio of the fluorescence image $[H_2O_2]_{cyto/mito}$ (490/420 nm) denoted as Rf. Rf values were normalized by precontraction Rf (R_{pre}). Values are expressed as means ± SE. A two-way ANOVA was used. *Significantly different from precontraction ($P < 0.05$); #significant difference between CONC and ECC for the same time points ($P < 0.05$). n = number of male mice.

difference between CONC and ECC. Although $[H_2O_2]_{cyto}$ levels increased after contractions in both contraction modes (CONC: 1.06 ± 0.02, ECC: 1.11 ± 0.01, $P < 0.0001$, main effect of time), the increase was significantly greater in ECC than CONC, with higher values maintained for 60 min. The differences in these profiles are presented in Fig. 3 (see next section). In the $[H_2O_2]_{mito}$ profile (Fig. 2, *bottom*), unlike $[H_2O_2]_{cyto}$, there were defined areas with high ratio values in the muscle fibers. Including these areas, the $[H_2O_2]_{mito}$ profile of the entire fiber was increased in both contraction modes ($P < 0.0001$, main effect of time), but there were no differences observed across contraction modes ($P = 0.496$). Interestingly, $[H_2O_2]_{mito}$ did not peak immediately after contractions for either CONC (i.e., peak delayed >20–25 min) or ECC (i.e., peak delayed 10–15 min).

In Vivo Cytosolic H_2O_2 Concentration Dynamics Profiles after Concentric and Eccentric Contractions

We performed a kinetics analysis to further elucidate the dynamics of H_2O_2 after muscle contractions. The mean dynamic $[H_2O_2]_{cyto}$ profiles for CONC and ECC are shown in Fig. 3*B*. The criterion correlation coefficients (r) for the goodness of fit of the models were not different between CONC ($r = 0.635 \pm 0.27$) and ECC ($r = 0.626 \pm 0.24$). R levels after contractions (indicating the change in $[H_2O_2]_{cyto}$ from precontraction) were higher for ECC (CONC: 1.06 ± 0.02, ECC: 1.11 ± 0.02, $P < 0.05$; Fig. 3*D1*). ΔR (decrease in $[H_2O_2]_{cyto}$ from postcontraction) was greater for CONC than ECC (CONC: −0.06 ± 0.01, ECC: 0.00 ± 0.02, $P < 0.05$; Fig. 3*D2*). Time delay (TD) was greater for ECC than CONC (CONC: 4.21 ± 1.73, ECC: 20.01 ± 4.89, $P < 0.05$; Fig. 3*D3*). Time constant (τ) did not differ between contraction modes (CONC: 7.25 ± 3.85, ECC: 4.51 ± 1.44, $P = 0.51$; Fig. 3*D4*).

Effect of NADPH Oxidase 2 Inhibitor (GSK2795039) on $[H_2O_2]_{cyto}$ before and after Eccentric Contractions of Tibialis Anterior Muscle

ECC increased $[H_2O_2]_{cyto}$ more than CONC. We focused on Nox2, which is localized at the plasma membrane and T-tubules and is a major source of cytosolic ROS produced by muscle contractions (12, 15). Figure 4 shows a representative ratio image of ECC after 20 min of exposure to GSK2795039, Nox2-specific inhibitor, on TA muscle. Twenty minutes of exposure to GSK2795039 did not change precontractions $[H_2O_2]_{cyto}$ (data not shown). The ~10% increase in $[H_2O_2]_{cyto}$ ratio values seen with ECC was abolished by the Nox2 inhibitor (ECC: 1.11 ± 0.02, $P < 0.05$ vs. precontraction, ECC + GSK: 1.02 ± 0.05, $P > 0.99$ vs. precontraction).

The Effects of Contraction Modes and H_2O_2 Levels on Gene Expression

The effects of contraction modes and H_2O_2 levels on gene expression were determined compared with the surgery-only noncontracted group. Analysis of muscle after 1 h

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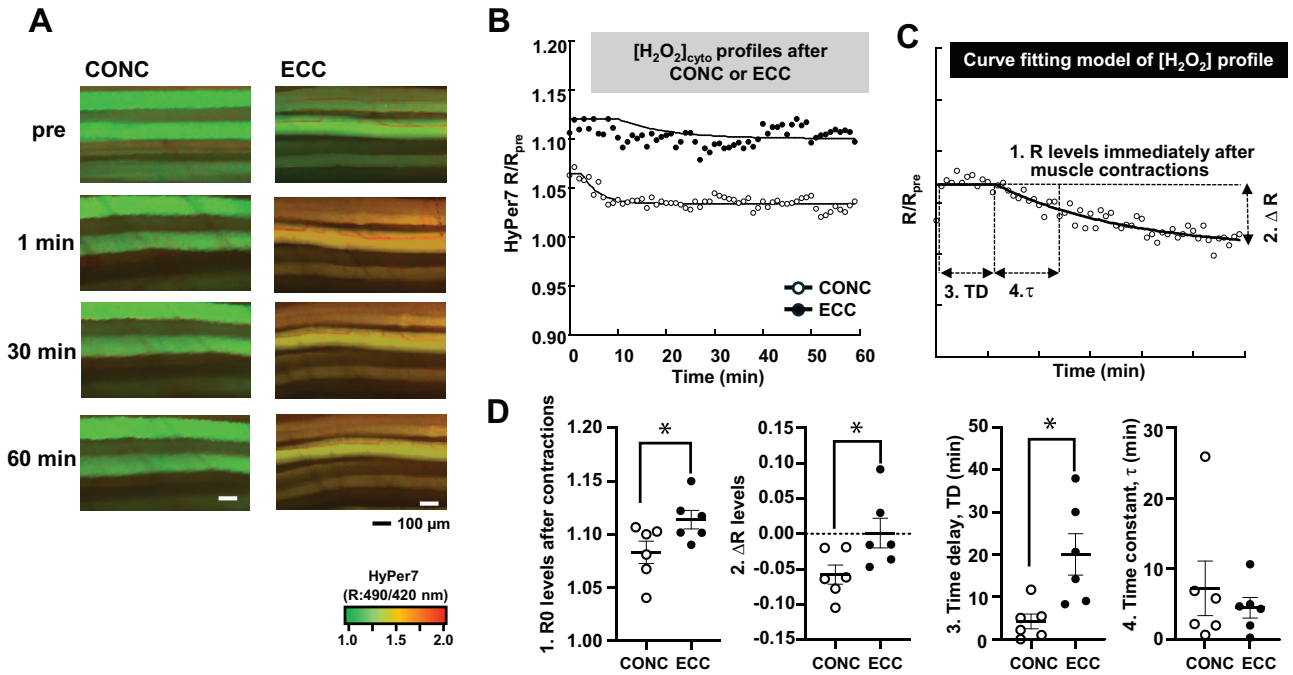


Figure 3. In vivo cytosolic hydrogen peroxide (H_2O_2) concentration ($[H_2O_2]_{cyto}$) kinetics profiles after concentric contractions (CONC) and eccentric contractions (ECC). *A*: typical HyPer7 fluorescence images after CONC and ECC are shown. HyPer7 fluorescence images were taken for 60 min postcontractions. Scale bar is 100 μm. *B*: plots showed the mean value for each protocol. Time-dependent change in the ratio of the fluorescence image $[H_2O_2]_{Cyto}$ (490/420 nm) denoted as Rf. Rf values were normalized by precontraction Rf (R_{pre}). *C*: schematic diagram showing the three parameters of the curve-fitting model. *D*: *1*) R0 represents R (=Rf/R_{pre}) level immediately after muscle contractions, *2*) ΔR represents the amplitude of each response, *3*) time delay (TD), and *4*) time constant (τ) represents the overall dynamics profile. Values are expressed as means ± SE (CONC: $n = 6$, ECC: $n = 6$, animals). Two-sided Student's *t* test was conducted for statistical significance. $^*P < 0.05$.

following contractions showed that 222 genes with q < 0.2 were present in the ECC group and none in the CONC and ECC + GSK groups. There were 41 upregulated genes whose expression increased more than 1.5-fold and 24 downregulated genes whose expression decreased to less than 0.67-fold compared with the sedentary group. The top 10 genes that differed between the sedentary and ECC groups were sorted by significance (i.e., q value) and fold-change (Tables 1 and 2, respectively). GO analysis of upregulated and downregulated genes revealed that GO terms were enriched only in upregulated genes. The enriched GO terms included blood vessel development, negative regulation of phosphorus metabolic process, cellular response to growth factor stimulus, and positive regulation of apoptotic processes (Fig. 5).

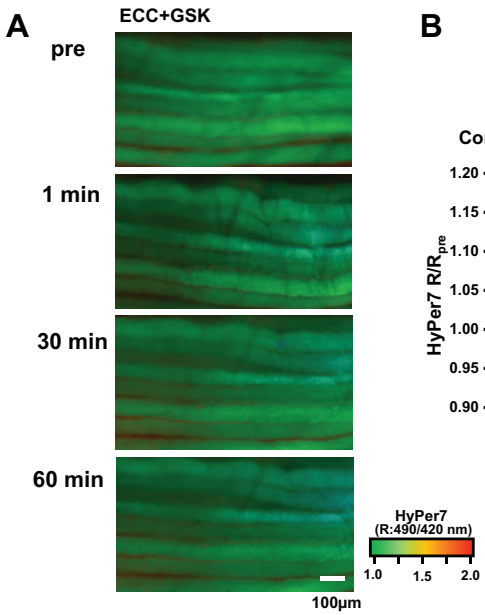


Figure 4. Effect of NADPH oxidase 2 (NOX2) inhibitor (GSK) on $[H_2O_2]_{cyto}$ before eccentric contractions (ECCs) of tibialis anterior muscle in vivo. *A*: HyPer7 fluorescence images were taken before and for 60 min after ECC + GSK. Scale bar is 100 μm. *B*: time-dependent change in the ratio of the fluorescence image $[H_2O_2]_{cyto}$ (490/420 nm) denoted as Rf. Rf values were normalized by precontraction Rf (R_{pre}). The ECC data are the same as presented in Fig. 2. Values are expressed as means ± SE. A two-way ANOVA was used. #Significant difference between ECC and ECC + GSK for the same time points ($P < 0.05$). n = number of male mice.

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Table 1. *Top 10 genes that are influenced by eccentric contractions sorted by fold-change*

Upregulated ECC	Downregulated ECC
Egr3 (2.7)	Gm9812 (−1.0)
Gm29233 (2.6)	Vwf (−0.9)
Fos (2.6)	Chodl (−0.9)
Atf3 (2.3)	Six5 (−0.8)
Egr1 (2.2)	Gja4 (−0.8)
Thbs1 (2.1)	Dipk2b (−0.8)
Socs3 (2.1)	Sncg (−0.7)
Serpine1 (2.1)	Kank3 (−0.7)
Fosb (2.1)	Ifi203-ps (−0.7)
Btg2 (2.0)	Pde4c (−0.6)

The numbers inside the parentheses represent the log2 fold-change relative to the control. ECC, eccentric contractions.

DISCUSSION

This investigation resolved the compartment-specific response of $[H_2O_2]$ to CONC and ECC in TA myocyte cytosol and mitochondria and, using comprehensive RNA analysis, related those responses to altered gene expression. The principal original findings are *1*) ECC increased $[H_2O_2]_{cyto}$ more than CONC, and this increase was sustained for at least 60 min into recovery after contractions. This $[H_2O_2]_{cyto}$ elevation by ECC was completely suppressed by Nox2 inhibition (GSK). *2*) $[H_2O_2]_{mito}$ did not differ by contraction modes, with both increasing slowly (10–25 min) after contractions had ceased. *3*) ECC upregulated genes included those related to blood vessel development, cellular responses to growth factor stimulus, and positive regulation of apoptotic processes, but no significantly altered genes were identified after CONC or ECC + GSK.

Concentric versus Eccentric Contractions and Cytosolic and Mitochondrial H_2O_2 Levels

Muscle contractions increase intramuscular ROS levels both in vivo and in vitro (30, 38, 39) but, to our knowledge, differences in intramuscular ROS levels between CONC and ECC contraction modes have not heretofore been identified. Among those few articles that have focused on changes in myocyte ROS levels across contraction modes, Kon et al. (28) measured the oxidative stress marker thiobarbituric acid reactive substances (TBARS) immediately after 12 min of CONC and ECC in TA from young mice. TBARS values increased for both CONC and ECC, but no differences were observed between the contraction modes. Sloboda et al. (40) isolated mouse lumbrical (LMB) muscles and found no difference in the increased fluorescence of fluorescent dichlorofluorescein (DCF), which is produced from dichlorodihydrofluorescein (DCFH) by ROS, across isometric and lengthening (ECC) contractions in vitro. These results differ markedly from the differences in $[H_2O_2]_{cyto}$ between CONC and ECC found in the present investigation, which is likely due to the methodological limitations in measuring ROS levels. Specifically, TBARS reacts with malondialdehyde (MDA), one of the end products of lipid peroxidation, resulting in an increased fluorescence and does not capture changes in H_2O_2. DCFH-DCF also has limitations when trying to quantify intracellular ROS levels in skeletal muscle because it is an irreversible probe that does not react directly with H_2O_2 and may, in and of itself, lead to de novo generation of H_2O_2 and other ROS (41, 42). It is also pertinent that, because the fluorescence of DCF is sensitive to local O_2 levels and pH, at the supraphysiological levels of hyperoxia typically used in vitro muscle/single myocyte preparations, it may be not able to detect any extant differences in $[H_2O_2]$ (4, 43–45). Typically, in predominantly fast-twitch murine muscles, perimyocyte O_2 partial pressures (PO_2) during contractions are <20 mmHg (46–48), which contrasts markedly with the 150 or 700 mmHg incident PO_2 using room air or 100% O_2. Thus, preservation of the in vivo environment in the present investigation and the use of HyPer7, a highly H_2O_2-sensitive fluorescent protein, may account for the ability to detect differences in H_2O_2 levels due to contraction modes herein, whereas previous studies could not.

Inhibition of the ECC-induced increase in $[H_2O_2]_{cyto}$, present after ECC but not CONC, by GSK2795039, a Nox2-specific inhibitor, provides strong evidence that activation of Nox2 is responsible for the ECC-induced increase in $[H_2O_2]_{cyto}$. Previous studies have reported that Nox2 is activated during muscle contraction in skeletal muscle (4, 12, 14, 15) and by stretch in cardiac muscle (49). Since ECC is a combination of muscle contraction and stretching, it is possible that ECC activates Nox2 more strongly than CONC. However, activation of Nox2 in mouse skeletal muscle returned to resting levels within 60 min after high-intensity interval running exercise more akin to CONC herein than ECC (14). Thus, the increase in $[H_2O_2]_{cyto}$ via sustained Nox2 activation may be ECC-specific. Our laboratory has previously shown, using a contraction protocol similar to the present investigation, that ECC causes a sustained cytosolic Ca^{2+} accumulation after ECC due to the influx of Ca^{2+} from the extracellular via stretch-activate channels (50). Ca^{2+} is known to activate Rac1, one of the subunits required for Nox2 activation (51). Therefore, the sustained increase in $[H_2O_2]_{cyto}$ after ECC may be due to Ca^{2+}-mediated activation of Nox2. In addition, the increase in Nox2-derived ROS by ECC may be necessary for the process of muscle regeneration from ECC-induced damage. In a previous study, Nox2-derived ROS were shown to be critical for gecko tail regeneration (52). It will be important to verify these relationships in future prospective investigations and, specifically, to resolve the effects of Nox2-derived ROS on muscle damage and regeneration processes after ECC.

$[H_2O_2]_{mito}$ showed a gradual increase after contraction with no detectable difference between contraction modes. In a study in which mitochondrial $O_2^{\bullet-}$ was measured in

Table 2. *Top 10 genes that are influenced by eccentric contractions sorted by significance*

Upregulated ECC	Downregulated ECC
Junos (1.0)	Chd3 (−0.6)
Fos (2.6)	Dipk2b (−0.8)
Thbs (2.1)	Mrvi1 (−0.6)
Nfkbiz (1.6)	Rsrp1 (−0.6)
Spry4 (0.9)	Chodl (−0.9)
Midn (0.7)	Fut11 (−0.6)
Ifrd1 (0.6)	Gm9812 (−1.0)
Gem (1.7)	Six5 (−0.8)
Ppp1r15a (0.9)	Ifi203-ps (−0.7)
Nr4a1 (0.7)	Gja4 (−0.8)

The numbers inside the parentheses represent the log2 fold-change relative to the control. ECC, eccentric contractions.

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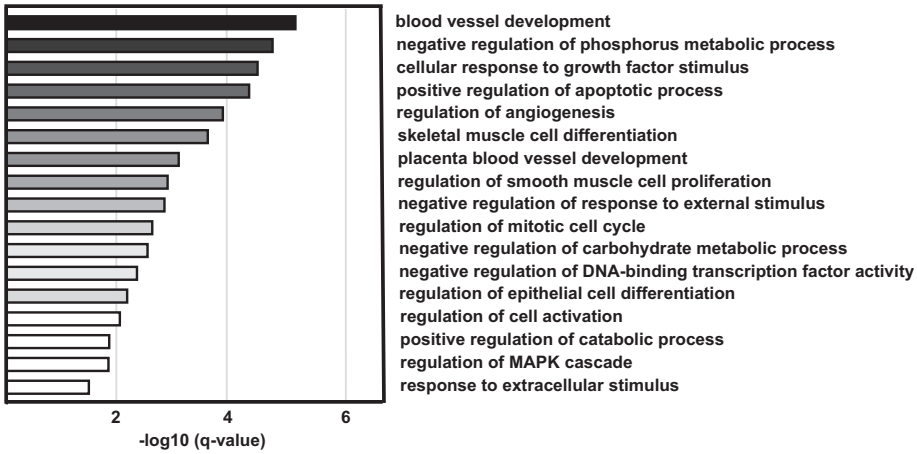


Figure 5. Significant GO terms for DEGs upregulated by ECC. DEGs, differentially expressed genes; ECC, eccentric contractions; GO, Gene Ontology.

isolated mouse flexor digitorum brevis (FDB) single muscle fibers during contractions, with incident O_2 levels not specified, using the mitochondrial ROS-sensitive fluorescent indicator Mito-sox, mitochondrial [$O_2^{\bullet-}$] slowly increased after CONC (30 V 10 min, duration for 0.5 s repeated every 5 s at 50 Hz) (13). In addition, a study in which Mito-roGFP2-Orp1, a mitochondrial H_2O_2-sensitive fluorescent protein, was expressed in single muscle fibers reported that $[H_2O_2]_{mito}$ was lower than at rest immediately after 15 min of concentric twitch contractions of mouse FDB (10 V, 1 Hz, 0.1 ms duration) (12). Furthermore, when C2C12 cells are stimulated to mimic metabolic changes during muscle contraction, $[H_2O_2]_{mito}$ are reported to be lower than at rest (15). Although in vivo/vitro and muscle contraction conditions differ, our results and those of previous studies suggest that mitochondrial ROS may decrease during contractions and gradually increase immediately after contractions, as proposed by Henriquez-Olguin et al. (8). In summary, these results demonstrated that H_2O_2 dynamics are different between intracellular compartments; H_2O_2 during muscle contraction appears to increase in the cytosol through Nox2, and H_2O_2 after contraction increases in the mitochondria.

Effect of Contraction Modes on Gene Expression

Changes in H_2O_2 levels due to muscle contractions activate redox signaling pathways and increase expression of multiple genes (3, 7). Compared with the CONC and ECC + GSK groups, the ECC group had higher $[H_2O_2]_{cyto}$ after contractions, and these high levels persisted for at least 60 min. Since the increase in H_2O_2 levels due to muscle contractions is associated with an increase, decrease, or fluctuation in the expression of various genes, we performed a comprehensive gene analysis using RNA-Seq, assuming that more genes were upregulated in the ECC group than in the other two groups. The results showed that DEGs were present only in the ECC group and wholly absent from the CONC and ECC + GSK groups. This indicates that the impact of ECC on gene expression is greater than that by CONC and supports that Nox2-derived ROS are involved.

GO analysis of upregulated genes in ECC revealed an enrichment of pathways related to vascular development, apoptosis-related muscle growth, muscle differentiation, and the MAPK pathway. ECC is known to induce apoptotic responses. Moreover, experiments in rat plantaris single-muscle fibers have shown a greater degree of activation of the MAPK pathway for ECC compared with CONC (53, 54). In human studies, pathways related to muscle differentiation were more enriched 4 h after ECC than CONC (35). Furthermore, the present data are consistent with the results of ROS-dependent activation of apoptotic pathways and MAPKs in myocytes (53, 55). The MAPK signaling pathway is associated with muscle protein synthesis and muscle remodeling (56), suggesting that H_2O_2 plays an important role in the regulation of muscle mass through muscle contractions. The relationship between $[H_2O_2]_{cyto}$ and the magnitude of change for genes with small q values (i.e., most of the genes whose expression was increased by ECC in the present investigation), such as FBJ osteosarcoma oncogene (*FOS*), activating transcription factor 3 (*ATF3*), and early growth response 3 (*EGR3*), is especially interesting as these genes are upregulated in colon cancer cell lines when exposed to H_2O_2 (57).

The expression of *Nfkbiz*, for example, first shown to be altered by ECC herein, may be related to ECC-induced skeletal muscle adaptation. IκB-ζ, the gene product of *Nfkbiz*, which regulates the expression of inflammatory cytokines by interacting with NF-κB, is itself a transcription factor that initiates inflammatory responses (58). Although the role of IκB-ζ has not been reported in skeletal muscle, we speculate that the gene expression of Nfkbiz is likely to be increased in ECC, given that NF-κB is involved in muscle damage caused by ECC (59) and that NF-κB is activated by H_2O_2 (60).

On the other hand, no DEGs were detected in ECC + GSK or CONC. This suggests that the high levels of ROS are involved in the expression of DEGs by ECC. Indeed, *Serpine1* gene expression increases with increasing concentrations of ROS; at least in breast cancer cells (61). However, it has also been reported that *Serpine1* and *Thbs1* (DEGs in this study) increase after CONC but not ECC (62). These observations support the concept that the DEGs in this study are also regulated by ROS-independent signaling and pathways, although their gene expression is upregulated by high ROS levels. For example, *Egr1* gene expression has been shown to be regulated by Ca^{2+} (a non-ROS signal) as well (63). Collectively, these findings are consistent with ROS being

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involved in ECC-induced changes in gene expression, but that these changes are not necessarily regulated solely by ROS-dependent pathways.

Limitations

A limitation of this study is that only male mice were used and no evaluation in female mice was carried out. It has been suggested that sex differences cause differences in ECC-induced muscle damage in rodents and humans and that ECC-induced muscle damage is less likely to occur in females than in males (4, 64–66). This difference may be related to the inhibitory effect of the female hormone estrogen on oxidative stress (67). Estrogen treatment has been shown to reduce the oxidative stress marker myeloperoxidase after exercise in male rats (68). Considering these findings, the persistence of higher H_2O_2 levels after ECC may not occur in females and differences in gene expression might potentially be observed. This unresolved and important aspect of sexual dimorphism needs to be researched in the future.

Summary and Perspectives

Using an in vivo bioimaging model that maintains blood circulation and a close-to-physiological muscle physicochemical environment, we have elucidated the $[H_2O_2]$ changes in the myocyte cytosolic and mitochondrial compartments across CONC and ECC contraction modes. Inhibition of Nox2 revealed that the attainment and maintenance of high $[H_2O_2]_{cyto}$ by ECC occur primarily via Nox2 and such inhibition also suppresses the gene expression changes after ECC. Because Nox2 and Nox4 are major sources of ROS production in skeletal muscle, further prospective quantitative and temporal investigations are needed to fully elucidate the regulatory potential of the Nox family. Given that ROS levels are involved in muscle damage, the increase in ROS levels via Nox2 by ECC is highly likely to be mechanistically implicated in muscle damage. Thus, the results of the present investigation provide important clues that help explain the mode-dependent differences in muscle damage and adaptation between ECC and CONC exercise.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL MATERIAL

Supplemental Figs. S1 and S2: https://doi.org/10.6084/m9.figshare.26156413.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

R.K. conceived and designed research; R.K. and H.S. performed experiments; R.K., T.K., R.T., and Y.K. analyzed data; R.K., D.C.P., Y.K., and D.H. interpreted results of experiments; R.K. prepared figures; R.K. drafted manuscript; R.K., R.T., H.S., D.C.P., Y.K., and D.H. edited and revised manuscript; R.K., T.K., R.T., H.S., D.C.P., Y.K., and D.H. approved final version of manuscript.

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